

Insight 3

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
%pwd
```

'/content/drive/MyDrive/BAIM - Sanjeev Jha'

```
%cd /content/drive/MyDrive/BAIM - Sanjeev Jha
```

/content/drive/MyDrive/BAIM - Sanjeev Jha

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
base_path = "/content/drive/MyDrive/BAIM - Sanjeev Jha/"

files = {
    "NVDA": base_path + "nvda.csv",
    "AMD": base_path + "amd.csv",
    "MU": base_path + "mu.csv",
    "TXN": base_path + "txn.csv",
    "SOXX": base_path + "SOXX.csv",
    "AVGO": base_path + "avgo.csv"
}
```

```
dfs = {}

for ticker, path in files.items():
    df = pd.read_csv(path)
    df['short_move'] = df['close'] - df['open']      # intraday price mo
    df['volatility'] = df['high'] - df['low']        # candle range
    dfs[ticker] = df

dfs['NVDA'].head()
```

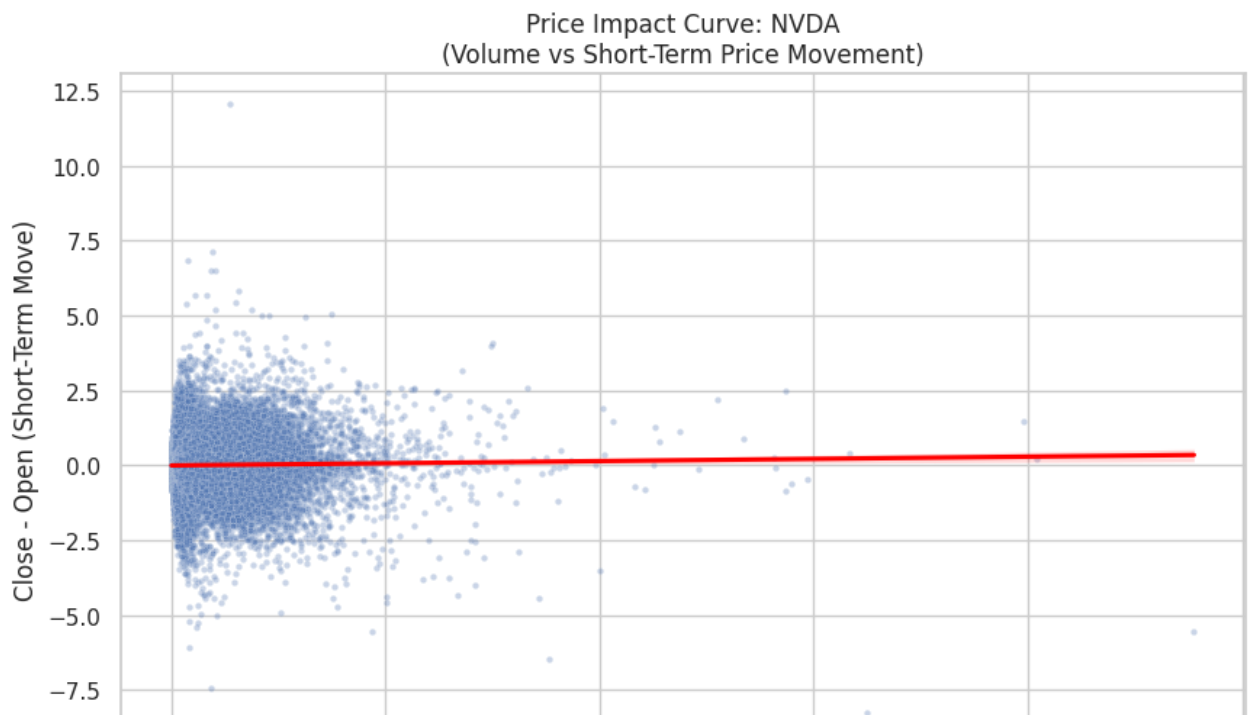
	tickerdate	tickertime	open	high	low	close	vw_price	volume
0	2021-01-04	09:30:00	524.17	524.20	518.99	519.13	522.6221	208089
1	2021-01-04	09:31:00	518.58	522.20	518.58	520.41	520.7398	37966
2	2021-01-04	09:32:00	520.15	521.40	519.00	519.60	520.0849	30017
3	2021-01-04	09:33:00	519.31	519.91	518.50	519.91	519.1552	37906
4	2021-01-04	09:34:00	519.88	521.12	519.03	520.27	520.0864	33790

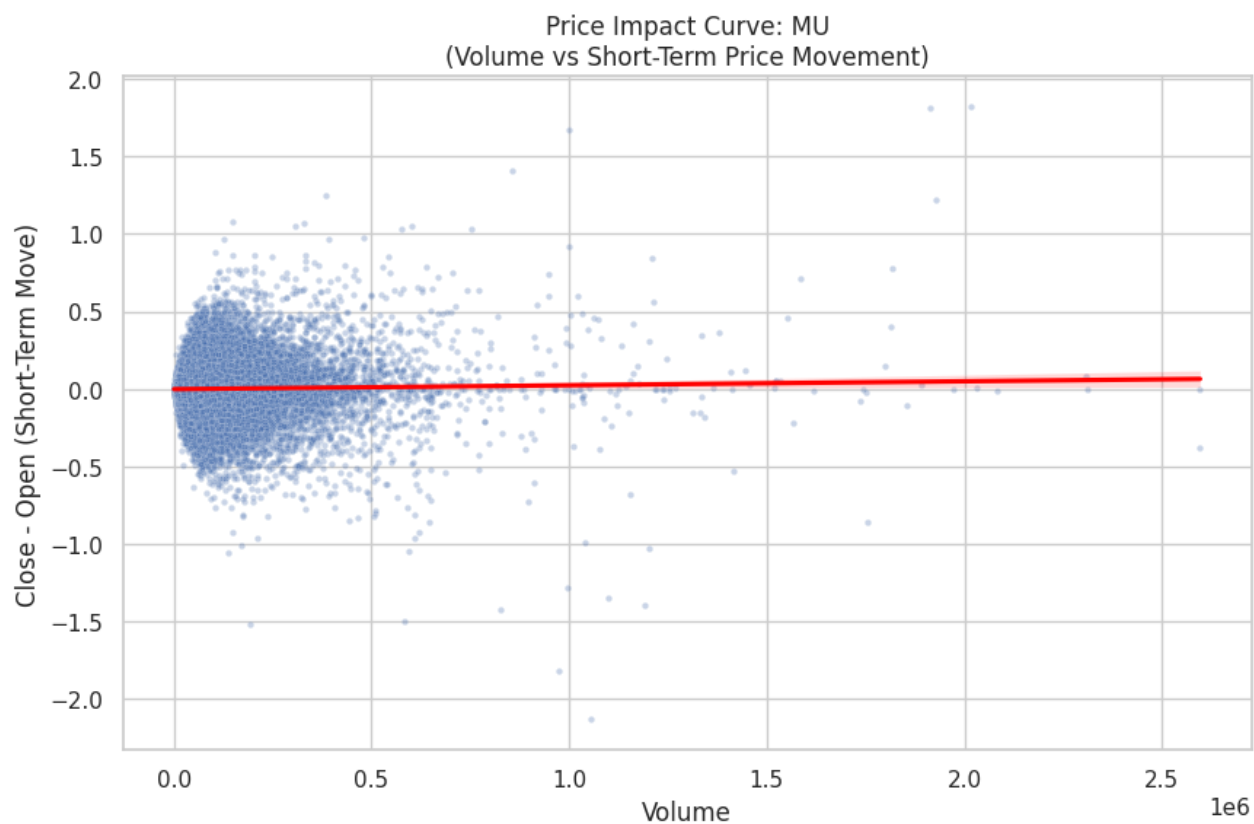
```
def plot_price_impact(df, ticker):
    plt.figure(figsize=(10,6))
    sns.scatterplot(
        x=df['volume'],
        y=df['short_move'],
        alpha=0.3,
        s=10
    )

    sns.regplot(
        x=df['volume'],
        y=df['short_move'],
        scatter=False,
        color='red'
    )

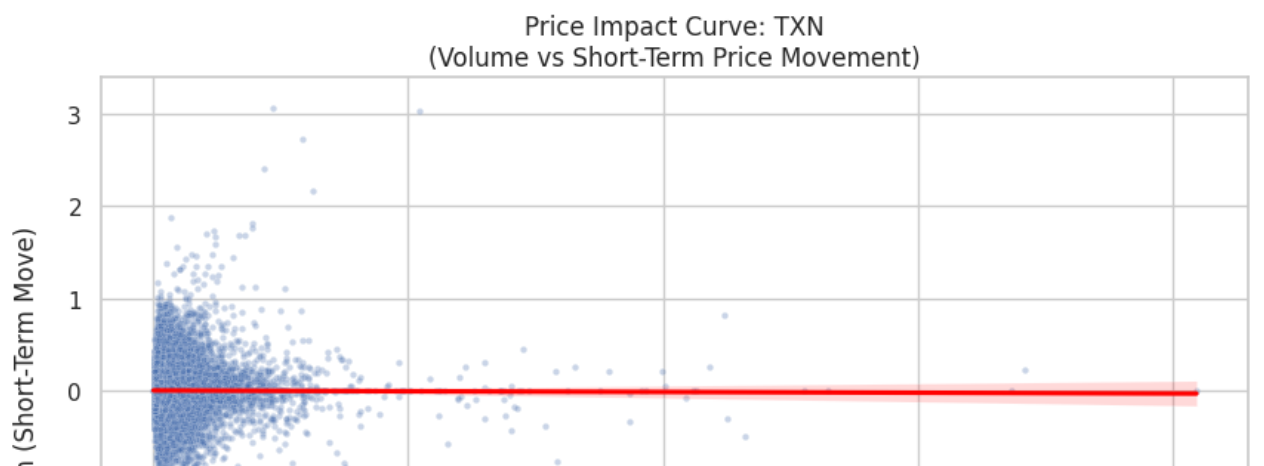
    plt.title(f"Price Impact Curve: {ticker}\n(Volume vs Short-Term P
    plt.xlabel("Volume")
    plt.ylabel("Close - Open (Short-Term Move)")
    plt.show()
```

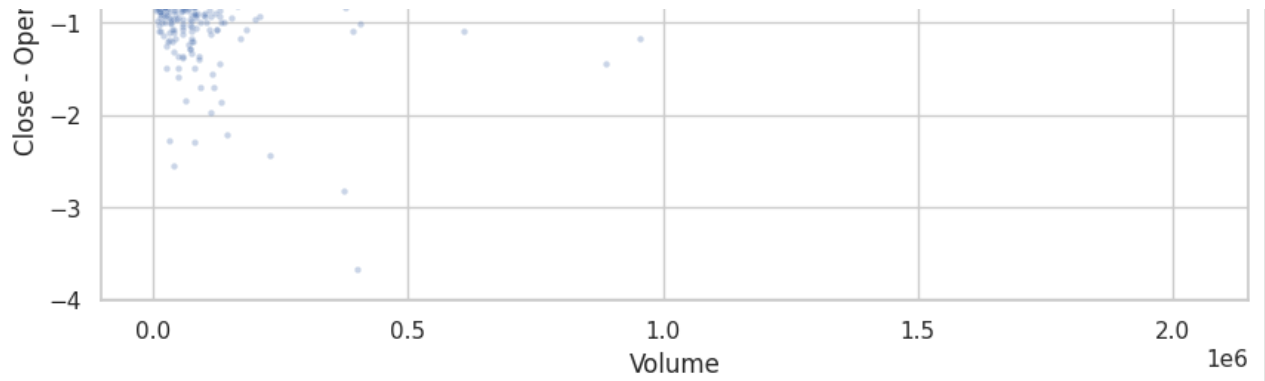
```
for ticker in ["NVDA", "AMD", "MU"]:
    plot_price_impact(dfs[ticker], ticker)
```



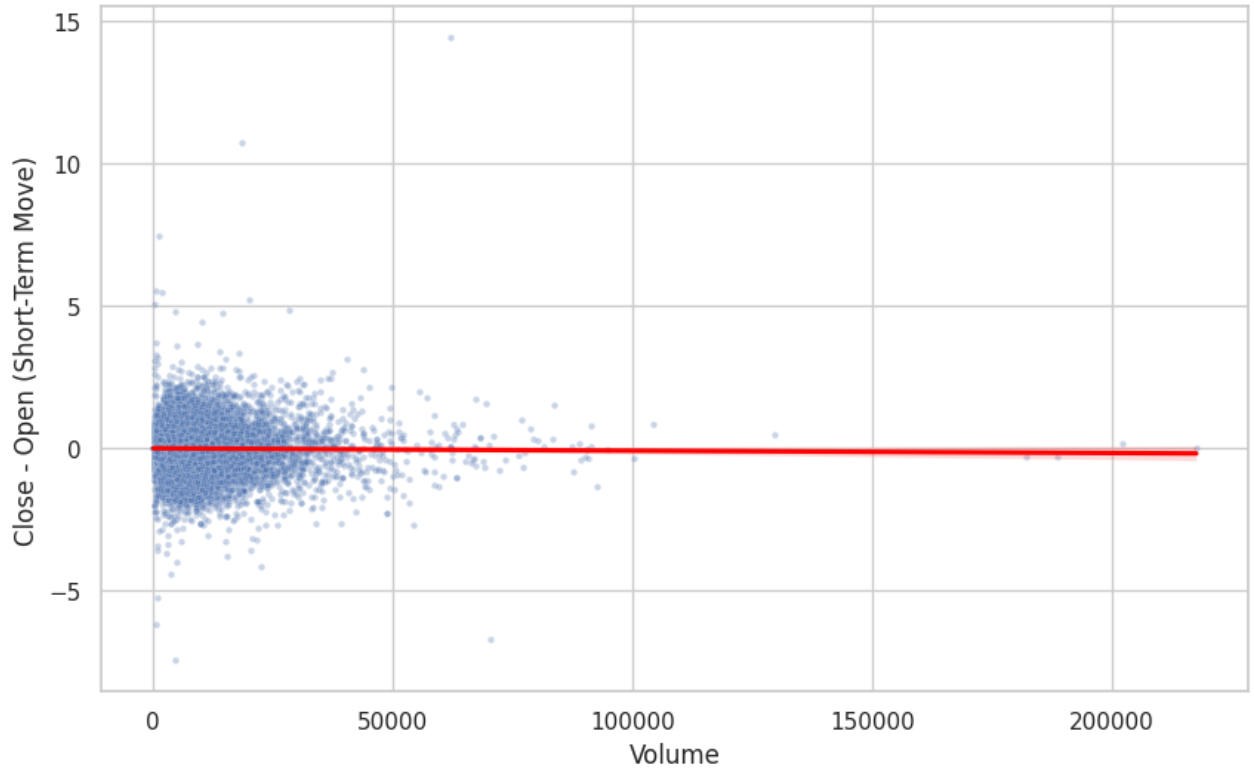


```
for ticker in ["TXN", "SOXX", "AVGO"]:  
    plot_price_impact(dfs[ticker], ticker)
```

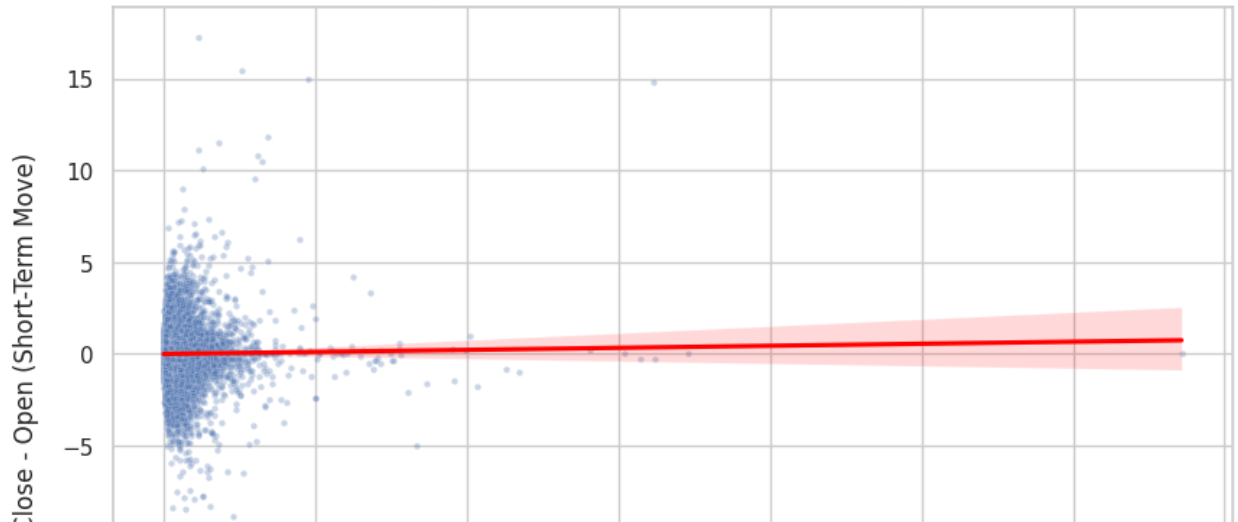


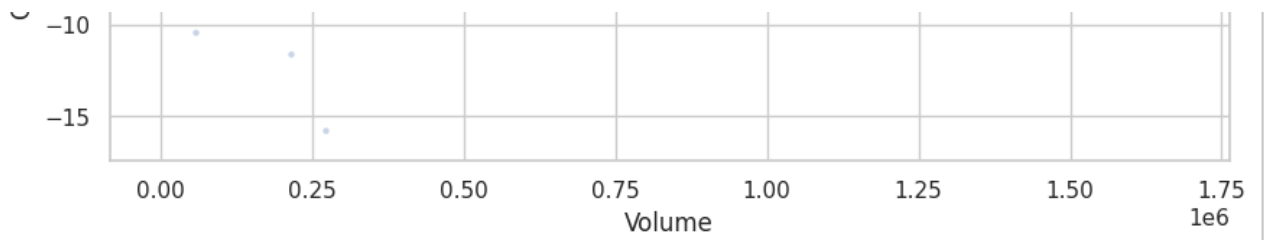


Price Impact Curve: SOXX
(Volume vs Short-Term Price Movement)



Price Impact Curve: AVGO
(Volume vs Short-Term Price Movement)





```
corr_results = {}

for ticker, df in dfs.items():
    corr = df[['volume', 'short_move']].corr().iloc[0,1]
    corr_results[ticker] = corr

corr_results
```

```
{'NVDA': np.float64(0.01687660429755416),
 'AMD': np.float64(0.036754361235878545),
 'MU': np.float64(0.01766606761686758),
 'TXN': np.float64(-0.0027945579578181564),
 'SOXX': np.float64(-0.009460668950504684),
 'AVGO': np.float64(0.009158801042814725)}
```



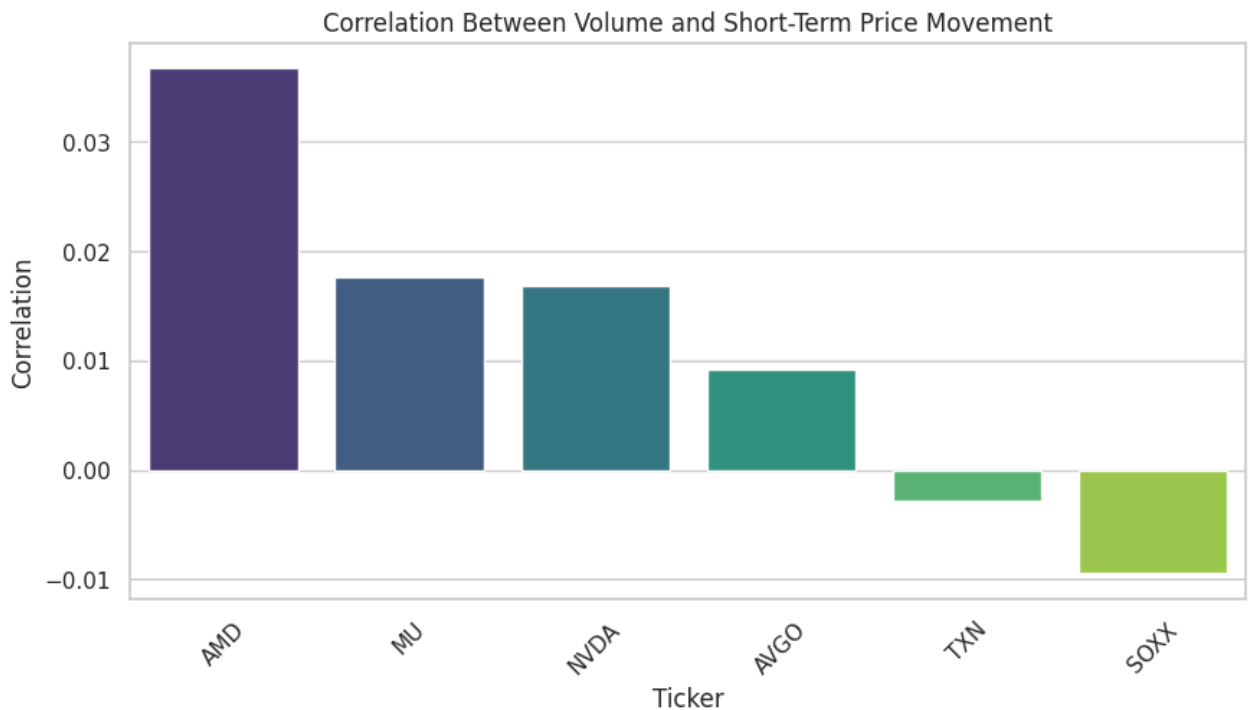
```
corr_df = pd.DataFrame.from_dict(corr_results, orient='index', columns=
corr_df.sort_values('correlation', ascending=False, inplace=True)

plt.figure(figsize=(10,5))
sns.barplot(x=corr_df.index, y=corr_df['correlation'], palette="virid
plt.title("Correlation Between Volume and Short-Term Price Movement")
plt.ylabel("Correlation")
plt.xlabel("Ticker")
plt.xticks(rotation=45)
plt.show()
```

/tmp/ipython-input-3791683556.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in a future version.

```
sns.barplot(x=corr_df.index, y=corr_df['correlation'], palette="virid
```



✓ Conclusive INSIGHT B

Market-Wide Volume & Price Movement

“Do semiconductor stocks move TOGETHER during volume surges?”

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
base_path = "/content/drive/MyDrive/BAIM - Sanjeev Jha/"

tickers = ["NVDA", "AMD", "MU", "TXN", "SOXX", "AVGO"]

files = {
    "NVDA": base_path + "nvda.csv",
    "AMD": base_path + "amd.csv",
    "MU": base_path + "mu.csv",
    "TXN": base_path + "txn.csv",
    "SOXX": base_path + "SOXX.csv",
    "AVGO": base_path + "avgo.csv"
}
```

```
dfs = {}

for ticker, path in files.items():
    df = pd.read_csv(path)

    # compute short-term movement & volatility
    df['short_move'] = df['close'] - df['open']
    df['volatility'] = df['high'] - df['low']

    dfs[ticker] = df

# preview NVDA
dfs["NVDA"].head()
```

	tickerdate	tickertime	open	high	low	close	vw_price	volume
0	2021-01-04	09:30:00	524.17	524.20	518.99	519.13	522.6221	208089
1	2021-01-04	09:31:00	518.58	522.20	518.58	520.41	520.7398	37966
2	2021-01-04	09:32:00	520.15	521.40	519.00	519.60	520.0849	30017
3	2021-01-04	09:33:00	519.31	519.91	518.50	519.91	519.1552	37906
4	2021-01-04	09:34:00	519.88	521.12	519.03	520.27	520.0864	33790

```
daily_data = []

for ticker, df in dfs.items():
    df['tickerdate'] = pd.to_datetime(df['tickerdate'])

    daily = df.groupby('tickerdate').agg({
        'volume': 'sum',
        'short_move': 'mean' # avg intraday movement
    }).reset_index()

    daily['ticker'] = ticker
    daily_data.append(daily)

daily_combined = pd.concat(daily_data, ignore_index=True)

daily_combined.head()
```

	tickerdate	volume	short_move	ticker
0	2021-01-04	12698725	-0.000870	NVDA
1	2021-01-05	6683883	0.037647	NVDA
2	2021-01-06	13194056	-0.063581	NVDA
3	2021-01-07	10514473	0.024041	NVDA
4	2021-01-08	6491149	-0.006215	NVDA

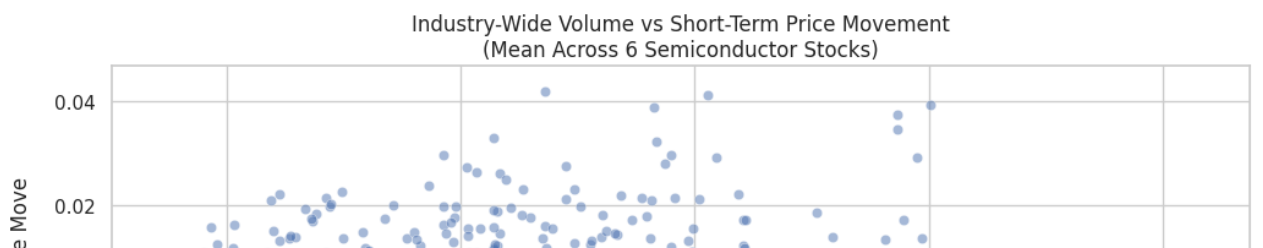
```
industry_daily = daily_combined.groupby("tickerdate").agg({
    "volume": "mean",
    "short_move": "mean"
}).reset_index()

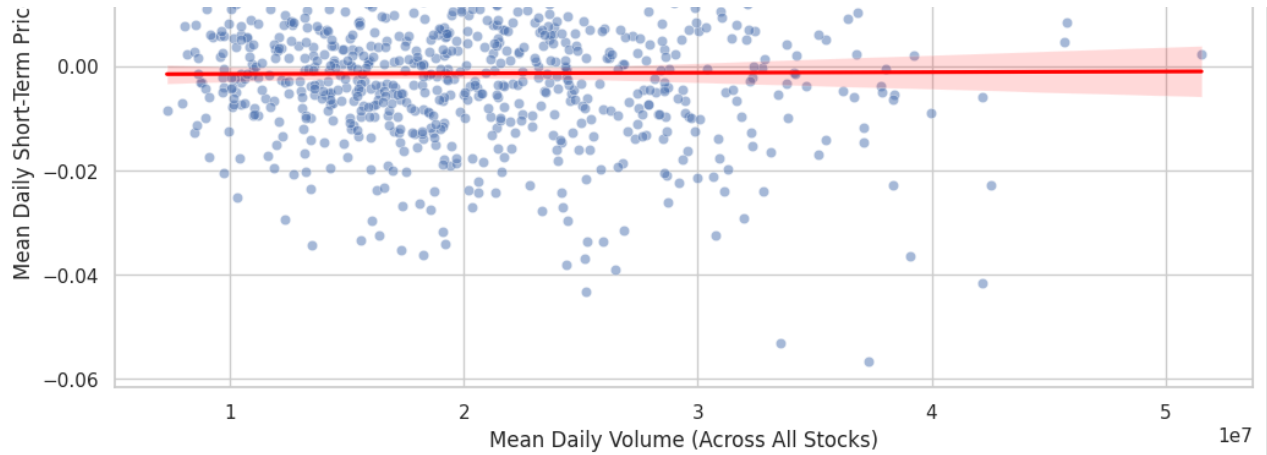
industry_daily.head()
```

	tickerdate	volume	short_move
0	2021-01-04	1.617749e+07	-0.009726
1	2021-01-05	1.224763e+07	0.022086
2	2021-01-06	1.469839e+07	-0.001671
3	2021-01-07	1.327979e+07	0.007325
4	2021-01-08	1.644758e+07	-0.003761

```
plt.figure(figsize=(12,6))
sns.scatterplot(
    x=industry_daily["volume"],
    y=industry_daily["short_move"],
    alpha=0.5
)
sns.regplot(
    x=industry_daily["volume"],
    y=industry_daily["short_move"],
    scatter=False,
    color="red"
)

plt.title("Industry-Wide Volume vs Short-Term Price Movement\n(Mean Across 6 Semiconductor Stocks)")
plt.xlabel("Mean Daily Volume (Across All Stocks)")
plt.ylabel("Mean Daily Short-Term Price Move")
plt.show()
```





```
industry_corr = industry_daily[["volume", "short_move"]].corr().iloc[0]  
industry_corr
```

```
np.float64(0.0066511644442437245)
```

INSIGHT C

Predictability of Price Impact

“Which stocks are most predictable based on volume?”

```
predictability = []

for ticker, df in dfs.items():
    corr = df[['volume', 'short_move']].corr().iloc[0,1]

    # regression slope: short_move = a + b*volume
    slope = np.polyfit(df['volume'], df['short_move'], 1)[0]

    predictability.append({
        "ticker": ticker,
        "correlation": corr,
        "slope": slope
    })

pred_df = pd.DataFrame(predictability)
pred_df
```

	ticker	correlation	slope
0	NVDA	0.016877	7.327296e-08
1	AMD	0.036754	3.479774e-08
2	MU	0.017666	2.542250e-08
3	TXN	-0.002795	-1.683530e-08
4	SOXX	-0.009461	-8.183410e-07
5	AVGO	0.009159	4.450512e-07

```
pred_df_sorted = pred_df.sort_values("correlation", ascending=False)
pred_df_sorted
```

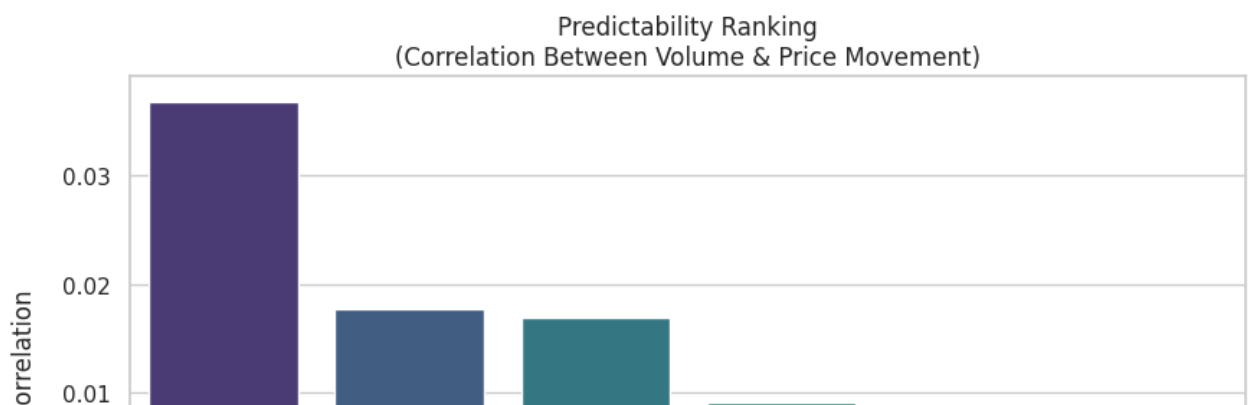
	ticker	correlation	slope
1	AMD	0.036754	3.479774e-08
2	MU	0.017666	2.542250e-08
0	NVDA	0.016877	7.327296e-08
5	AVGO	0.009159	4.450512e-07
3	TXN	-0.002795	-1.683530e-08
4	SOXX	-0.009461	-8.183410e-07

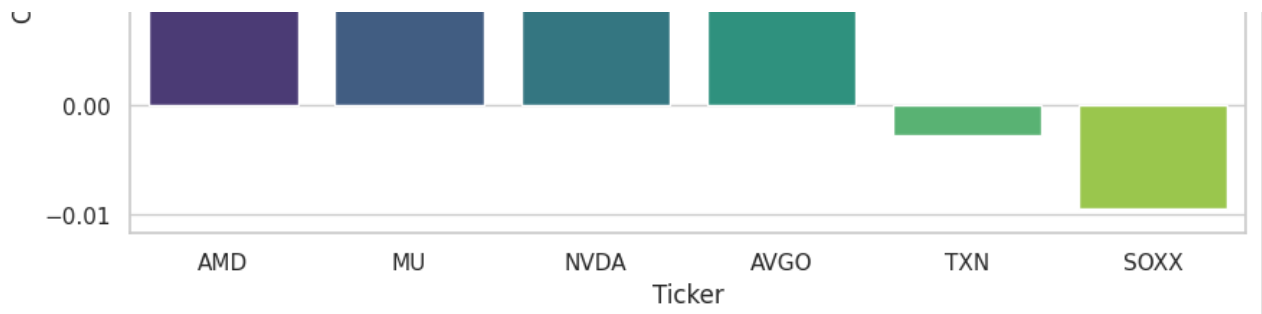
```
plt.figure(figsize=(10,5))
sns.barplot(
    x="ticker",
    y="correlation",
    data=pred_df_sorted,
    palette="viridis"
)
plt.title("Predictability Ranking\n(Correlation Between Volume & Price)
plt.ylabel("Correlation")
plt.xlabel("Ticker")
plt.show()
```

/tmp/ipython-input-4294426163.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in a future version. Use `color` or `hue` instead.

```
sns.barplot(
```

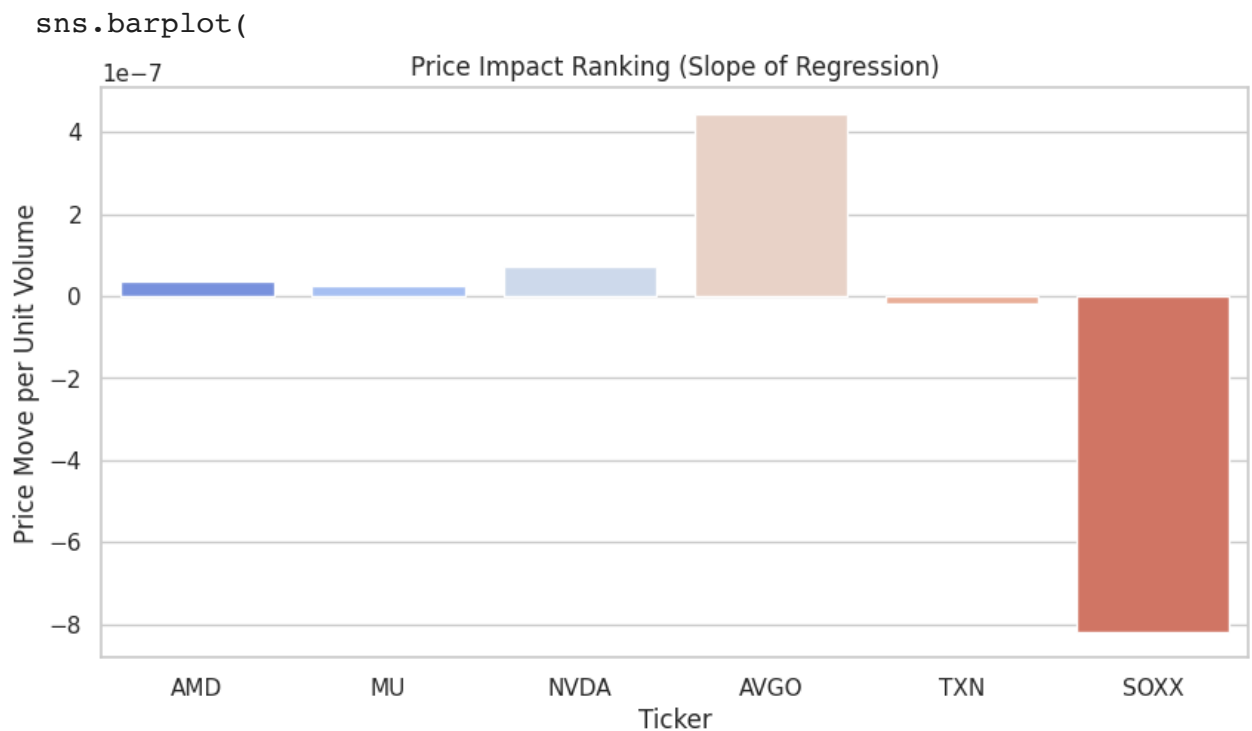




```
plt.figure(figsize=(10,5))
sns.barplot(
    x="ticker",
    y="slope",
    data=pred_df_sorted,
    palette="coolwarm"
)
plt.title("Price Impact Ranking (Slope of Regression)")
plt.ylabel("Price Move per Unit Volume")
plt.xlabel("Ticker")
plt.show()
```

/tmp/ipython-input-3657674644.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in a future version.



Start coding or generate with AI.